

RF receiver module (RX1), regulator 7805 (IC4), decoder HT12D (IC5), microcontroller ATmega16 (IC6), motor driver L293D (IC7) and a few discrete components.

Microcontroller ATmega16 is the heart of the circuit. ATmega16 is a low-power CMOS 8-bit microcontroller based on the AVR-enhanced RISC architecture. It has 16 kB of in-system programmable flash program memory, 512 bytes of EEPROM, 1kB SRAM, 32 general-purpose input/output (I/O) lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a byte-oriented two-wire serial interface, an 8-channel 10-bit analogue-to-digital converter, an SPI serial port and six software-selectable power-saving modes.

Data transmitted from the wireless controller is received by the RF receiver module and decoded by IC5. Data reception is indicated by LED6. Decoded data is fed to port pins PB0 through PB3 of microcontroller IC6. Identical addresses for the encoder and the decoder can be selected through switches DIP1 and DIP2. The outputs from pins PD0 through PD3 of the microcontroller are fed to IN1 through IN4 of motor-driver IC7 to drive motors M1 and M2, respectively. Outputs OUT1 and OUT2 drive motor M1, and outputs OUT3 and OUT4 drive motor M2. Enable pins (en1 pin 1) and EN2 (pin 9) are connected to port pins PD4 and PD5 of microcontroller IC6, respectively.

Regulator IC4 provides regulated 5V supply to the circuit. 12V battery supply is directly connected to pin 8 of motor driver IC7. Switch S3 is used for manual reset.

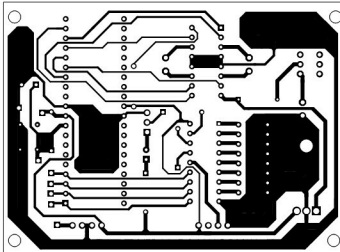


Fig. 7: An actual-size, single-side PCB for the robot

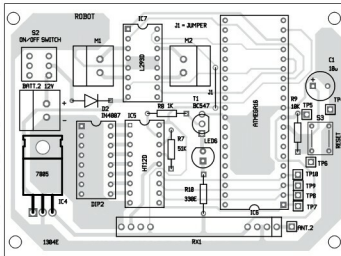


Fig. 8: Component layout for the PCB in Fig. 7

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

Construction and testing

Actual-size, single-side PCBs of wireless controller and robot are shown in Figs 5 and 7 and their component layouts in Figs 6 and 8, respectively. Assemble the circuit on a PCB as it saves time and minimises assembly errors. Carefully assemble the components and double-check for any overlooked error. Always use IC bases for ICs. Before inserting the microcontroller

and other ICs, always check the correct supply voltage. Suitable connectors are provided on both the PCBs to connect geared motors and power supplies.

Hold the wireless controller in your hand and tilt it to move the robot in the desired direction. Controller's sensitivity can be adjusted using VR1 through VR4.

Align the controller in your hand such that when you tilt the remote control down in front (forward), both motors M1 and M2 rotate in forward direction (clock-wise). When you tilt the remote control up in front (backward), both motors M1 and M2 should rotate in reverse direction (anti-clockwise). Both the circuits are simple and will immediately work after proper assembly. In case of any difficulty refer the test point table and verify the voltage levels.

Software

The source program is written in 'C' language and compiled using AVR Studio to generate the Intel hex code. The generated hex code is programmed using a suitable programmer. It is well commented and easy to understand.

The device is shipped with CKSEL = "0001" and SUT = "10." The default clock source setting is therefore 1MHz internal R-C oscillator with the longest startup time. To activate the internal oscillator, fuse bytes of the microcontroller have to be programmed as follows:

Fuse low byte = D4

Fuse high byte = 99 ●

The author is a second year B.Tech in electrical engineering from College of Engineering and Technology, Bhubaneswar. He likes working with electronic projects including microcontrollers